can you please carefully reason through the conversion to dB HL? specifically, i'm interested in sorting out:

- exactly how it should be implemented --> as previously discussed, we're already doing the conversion from dB FS to dB SPL for each pair of headphones, but it seems that the ISO standards around dB HL also include this PLUS the final bit to convert to HL. I would like to be able to ONLY implement the SPL to HL conversion as an additional option at the end of the chain, but it seems that the relevant ISO standards are not designed in such a way to work independent of headphones. is there perhaps a way to just us standard equal loudness curves instead?
- are there downsstream effects that aren't properly implemented? specifically, issues with plotting, reporting, setting thresholds, etc in PTA (and by extension in IFMC/TMC)?

Show less

Excavated audiology standards and code architecture implications >

I'll treat this as an analysis session — reading the current project snapshot to reason about the design and check downstream code, not editing anything yet. Before we make actual edits I'll want fresh uploads of your local copies (the `/mnt/project/` snapshot can lag). Flagging that up front, then digging in.

Let me ground the downstream review in the actual code rather than memory.

Validated converter design and investigated calibration configuration integration >

Clean, source-agnostic converter — correct sign convention, log-freq interp, extrapolation warns. Now where does the RETSPL table come from, and how does it get wired in?

Identified architectural discrepancies and mapped conversion implementation pathways >

Important: this snapshot shows SPL at 95.3/95.3 across all four profiles (placeholder), which differs from the 92.6L/91.4R DT250-A figure I have from your 2026-06-05 measurement. That's the `/mnt/project/` lag I mentioned — I'll trust your local copy on the actual numbers. Doesn't affect the HL reasoning; what matters is the structure, which is sound: **RETSPL lives per-profile in the calibration config, with a source label and a "null disables HL" convention.** That's the right home for it.

Let me trace the wiring and unit handling across the engine.

Traced conversion architecture across codebase to verify implementation >

Let me read the actual conversion points — input (HL→SPL), the summary-column write, and the plot path.

Verified conversion logic and traced downstream implementation pathways >

`_resolve_level` is correct: SPL passthrough, else `hl_to_spl(value, fc)`, with step sizes explicitly excluded (a dB step is unit-invariant — right call) and sensible init-deferral. Now the output side — summary column and plot.

Investigated HL conversion implementation and downstream effects across PTA systems >

`_threshold_to_hl` is defensive and gated on `level_units == "dB SPL"` + defined `fc` — which means for any paradigm not reporting SPL it silently writes no HL column. That's actually the right behavior for IFMC/TMC (more below). Let me confirm what IFMC/TMC report as units and `fc`, check the plot path, and find the safety-ceiling clamp.

Examined HL conversion implementation, identified latent fragility in IFMC/TMC, verified safety mechanis... >

Key findings so far:

- PTA: `level_units = "dB SPL"`, single `fc` → HL summary column written correctly.
- IFMC/TMC: report `masker_level_units` / `probe_level_units` but **no top-level `level_units`** → `_threshold_to_hl` returns None → no HL column. Correct outcome, but *accidental* — it hangs on IFMC/TMC simply not defining `level_units`. Flagging that as latent fragility.
- Safety ceiling is unit-safe: staircase + clamp both run in SPL, HL entries are converted to SPL *before* the clamp. Defense-in-depth in both `paradigm_pta` and `spl_to_amplitude`.

Now the plot/report path in the GUI.

Dissected HL conversion architecture and interrogated equal-loudness feasibility >

Read the wiring end-to-end. Summary up front, then the conceptual core (where your equal-loudness intuition needs pushback), then the grounded downstream review.

The architecture you want already exists — and it's correct

You asked for SPL→HL as a single optional final step, decoupled from the per-headphone SPL cal. That is exactly what's implemented: `threshold_dB` (SPL) is canonical/stored, and HL is a pure final offset `dB HL = dB SPL − RETSPL(f)` applied by `HLConverter`, with the RETSPL table living *per-profile* in `calibration_config.json`. The converter imports nothing from the rest of the toolbox, sign convention is right, log-freq interpolation, extrapolation warns. Nothing about the design needs to change.

What you *can't* have is a transducer-independent RETSPL. That's not an ISO drafting quirk you can route around — it's physics, and equal-loudness curves don't escape it.

Why equal-loudness / free-field curves don't help

RETSPL bundles two things:

1. the normal-hearing threshold expressed at the eardrum (frequency-dependent, roughly transducer-independent), and
2. the real-ear-vs-coupler coupling for *that specific earphone* (frequency- and earphone-dependent — RECD territory: seal, load impedance, leakage, placement).

Part (2) is the entire reason RETSPL is transducer-specific. Now consider the substitutes:

- **ISO 226 / MAF (equal-loudness, free-field threshold):** free-field, frontal-incidence, binaural. It bakes in the HRTF — head diffraction, the ~2–4 kHz concha/ear-canal resonance gain (15–20 dB) — all of which headphones *bypass*. To use a free-field curve with headphones you must add back the headphone-to-eardrum-vs-free-field-to-eardrum transfer, which is precisely the earphone-specific coupling (part 2) you were trying to avoid. The problem just relocates; it doesn't vanish.
- **MAP (minimum audible pressure, eardrum-referenced):** closer to your situation since your coupler SPL ≈ eardrum-plane SPL. But MAP curves are monaural-vs-binaural-confounded (the famous "missing 6 dB"), noisy at low frequency (physiological noise in a closed coupler), and not standardized the way RETSPL is. And they'd *still* need a coupler-vs-median-real-ear correction unless your coupler perfectly mimics the median ear (it doesn't — that's what RECD exists for).

So the "transducer independence" you want is the one thing the geometry won't give you for free.

Your B&K 4153 makes the cheap upgrade obvious

The relevant detail: ISO 389-8 (circumaural, HDA200) specifies the IEC 60318-1 artificial ear — which is the 4153 family. Your dB FS→SPL cal is measured *in that coupler*. So `dB HL = coupler_SPL − RETSPL_HDA200(f)` is dimensionally consistent, and the *only* residual error is DT250-vs-HDA200 (two circumaural phones), i.e. the ΔRECD term.

Compare your current placeholder: ISO 389-1 **TDH-39** is supra-aural in a 6cc coupler — a *double* mismatch (wrong earphone type *and* wrong coupler). Swapping TDH-39 → ISO 389-8 HDA200 keeps the architecture identical, just a better table, and collapses the error to one bounded, circumaural-to-circumaural dimension.

Two realistic endpoints for the table:

- **(B) ISO 389-8 HDA200** — traceable, immediate, residual error = ΔRECD(DT250,HDA200). Good enough for research display; flag the source.
- **(A) Biological calibration** — run a panel of otologically normal ears on the actual DT250+UCX II chain; median threshold per frequency *is* your 0 dB HL by construction. Correct by definition, but needs N≈10+ ears (between-subject SD is ±6–10 dB, so a "small pilot" tells you almost nothing).

One trap to rule out explicitly: measuring DT250 **and** HDA200 on the coupler only gives you their FR difference, not ΔRECD. ΔRECD is a real-ear-vs-coupler quantity — invisible to coupler-only measurement. So the cheap "measure both on the 4153" idea does **not** recover the correction; don't spend time on it. It's B or A.

Downstream review (against the project snapshot — verify against your local copy)

Correctly handled:

- **Input levels:** `_resolve_level()` converts start/min/max from HL→SPL at `self.fc`; step sizes are correctly *not* converted (a dB step is unit-invariant). Init-deferral is sane.
- **Safety ceiling:** unit-safe. HL entries are converted to SPL *before* the staircase, and `max_spl` clamps in SPL — both in `paradigm_pta` (L100–104) and again in `spl_to_amplitude` (L432). Defense-in-depth.

- **Threshold summary:** stores `threshold_dB` (SPL) *and* derived `threshold_dB_HL`; reversible; `_threshold_to_hl` never raises.
- **Provenance:** metadata header records `retspl_source`. Good — an HL value with no stated RETSPL is not defensible, and you're recording it.
- **Fit:** runs on SPL levels (`levels_dB`), consistent unit. HL is not in the math — matches the standing conclusion that HL is not the cause of the curve_fit runaway.
- **GUI:** `_operator_in_hl()` gates on both config *and* `hl_calibrated`; honest "HL unavailable, showing dB SPL" fallback rather than silent lying.

Latent / worth flagging:

1. **IFMC/TMC HL suppression is accidental.** HL is correctly *not* written for them — but only because they don't define a top-level `level_units` key (they have `masker_level_units`/`probe_level_units`), so `_threshold_to_hl`'s `level_units != "dB SPL"` gate returns None. If anyone later adds `level_units: "dB SPL"` for consistency, a wrong/ambiguous HL column appears — and `self.fc` for those paradigms is ambiguous (probe freq? masker freq?). Conceptually HL is a *category error* for IFMC/TMC anyway: those are masked / temporal-masking thresholds, not pure-tone detection at audiometric frequencies. SPL (or dB SL re the probe's own 16 ms threshold) is the right unit there. Recommend making the suppression *intentional* — an explicit `hl_reportable` opt-in flag rather than inferring from `level_units`.
2. **The "Previous Thresholds" plot is not an audiogram.** x = config_num (not log-frequency), y not inverted. It relabels correctly to "Threshold (dB HL)" in HL mode, but if you intend clinical audiogram semantics (inverted y, log-freq x, symbols) in HL mode, that's not implemented. Design-intent question, not a bug.
3. **SPL-mode y-label is "Threshold (dB)"** — ambiguous; should be "dB SPL" since the stored value is SPL.
4. **HL ceiling "moves" with frequency.** If the operator sets max in HL, the effective HL ceiling drops wherever `HL_max + RETSPL(f) > max_spl` (e.g. ~89.5 dB HL at 6 kHz with the current table). Physically correct, but surprising — consider warning when an HL max is clamped by `max_spl`.
5. **Extrapolation warnings** fire for any tested freq outside 125–8000 Hz. Fine for PTA; another reason HL shouldn't reach IFMC/TMC high-freq probes.

Net: the SPL/HL split is implemented the way you wanted and is sound. The real open work is the **RETSPL table choice** (a science decision, not a code one) and making **IFMC/TMC HL suppression explicit** rather than incidental.

Unresolved questions

1. RETSPL table: ship **ISO 389-8 HDA200** as the labeled research-grade approximation now, or commit to **biological cal** on the DT250 chain (and accept the panel-size cost)? Or both — 389-8 now, biological later?
2. Do you want HL *formally disabled* for IFMC/TMC via an explicit `hl_reportable=False`, or leave the incidental suppression (and accept the fragility)?
3. Is the threshold plot meant to become a real audiogram (inverted-y, log-freq-x) in HL mode, or stay a per-config QC scatter?
4. RETSPL per-profile vs shared: keep a copy in each profile (current), or reference one shared `retspl_*.csv` so the HDA200→DT250 upgrade is a one-file change across all profiles?

5. Source of the HDA200 numbers if we go (B) — ISO 389-8 table directly, or your transducer vendor's coupler data?

Nothing was edited this session, so there's nothing to push back to the project yet. Before we implement any of the above I'll want fresh uploads of your local working copies — the snapshot here already disagrees with your 2026-06-05 SPL numbers (it shows 95.3/95.3 placeholders), so it's lagging.

what exactly is ΔRECD?

Derived ΔRECD definition and connected it to RETSPL error >

RECD = Real-Ear-to-Coupler Difference. For a single transducer at a fixed drive, it's the dB difference, as a function of frequency, between the SPL that transducer produces at a real eardrum and the SPL it produces in a standardized coupler:

$RECD_E(f) = SPL_{realear,E}(f) - SPL_{coupler,E}(f)$ (same input drive)

It's nonzero because a real ear canal + eardrum impedance is a different acoustic load than the fixed coupler cavity, so the same source delivers different SPL to each. (It's the same quantity hearing-aid fitting measures per-patient, because residual canal volume varies; here the "ear" is the median normal ear.)

ΔRECD is just the difference of two transducers' RECDs on that same median-normal ear:

$\Delta RECD(f) = RECD_{DT250}(f) - RECD_{HDA200}(f)$

Why it's exactly the error term. At median-normal threshold with earphone E, the eardrum sees threshold pressure $P_{thr}(f)$ — a property of the auditory system, not the earphone — and the coupler reads $RETSPL_E(f)$ by definition. So $RECD_E = P_{thr} - RETSPL_E$, i.e. $RETSPL_E = P_{thr} - RECD_E$. P_{thr} cancels between earphones:

$RETSPL_{DT250} - RETSPL_{HDA200} = -(RECD_{DT250} - RECD_{HDA200}) = -\Delta RECD$

So if you compute HL with the HDA200 table on DT250 data, $HL_{true} = HL_{est} + \Delta RECD$, frequency by frequency. The entire transducer-specificity of RETSPL lives in $RECD_E$; the part that differs between your two phones is ΔRECD, and that *is* the bias.

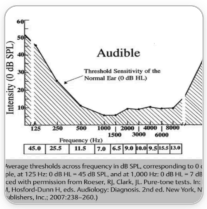
Why coupler-only measurement can't recover it. ΔRECD contains two real-ear terms ($SPL_{realear}$ for each phone) a coupler never sees. Model each earphone as a Thévenin source — open-circuit pressure p_E, source impedance Z_E — into a load Z: $SPL = p_E \cdot Z/(Z_E+Z)$. For one earphone, the real-ear/coupler ratio is $[Z_{ear}/(Z_E+Z_{ear})] / [Z_{coup}/(Z_E+Z_{coup})]$, and p_E cancels. So RECD depends only on the source impedance Z_E and the two loads — *not* on the earphone's frequency response p_E. A coupler FR measurement is dominated by p_E (the thing that cancels out of RECD) and gives you no real-ear load comparison, so it can't isolate ΔRECD. Recovering it needs probe-tube SPL in actual ears with each phone — nearly the biological-cal effort.

Expected magnitude. ΔRECD → 0 where the two phones have similar source impedance and seal/cavity geometry — broadly true mid-band for two well-sealing circumaural dynamics. It grows at the extremes: low frequency (leakage/seal, where the real-ear seal departs from the coupler seal differently per cushion) and high frequency (canal standing waves, placement sensitivity, cushion-cavity resonances). For your specific DT250-vs-HDA200 pair it's unmeasured — plausibly ~1–2 dB through the mids, more at the edges, but treat that as a physical expectation, not a number.

the HDA200 is no longer manufactured. i have been able to find some alternatives that should be work as a direct replacement...

"RadioEar DD450: This is the designated, official successor to the HDA 200. Built with similar passive ambient noise-attenuating earcups (like 3M Peltor cups), it covers a massive frequency range (up to 20 kHz for high-frequency audiometry) and has matching RETSPL parameters."

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that said, they are expensive and we're already delayed, so i want to sort this out ASAP with what we have already. we're not doing clinical diagnoses, so i think we can aim for "good enough" as long as we disclose the assumptions we make along the way - as you said: "(B) ISO 389-8 HDA200 — traceable, immediate, residual error = ΔRECD(DT250,HDA200). Good enough for research display; flag the source."

where i'm still a bit confused is that so far we've always just been trusting that the artificial ear is a "good enough" estimation of what a normal human ear hears, after all, that's the whole point, no? so why is it not sufficient to use this graph (the one i derived the current RETSPL data from)? we're already making the assumption that those SPL values are real(istic), i.e. including all real-world issues like headphone coupling to the head, ear canal impedance, etc., so it seems that using the curve as is would just be continuing the same assumptions

Show less

Untangled transducer-specific coupling assumptions from universal coupler principles >

You've put your finger on exactly the right question, and the kernel of your intuition is correct — but there's a specific place it breaks, and it's worth being precise about it. Let me get the exact replacement numbers first, then explain.

Reconciled conceptual confusion between coupler approximation and transducer-specific RETSPL values >

Got the replacement numbers and they make the case concrete. But first the conceptual knot, because you're half-right and the half that's wrong is the important half.

The graph is not "ear + coupler." It's "ear + coupler + TDH-39."

Your mental model is that the curve represents *the threshold of the normal ear as seen by the artificial ear* — a property of (ear, coupler). It isn't. It's the coupler SPL that **one specific earphone, the TDH-39, produces when a normal listener is at threshold**. The RETSPL value is the dB SPL that corresponds to 0 dB HL for a given headphone/coupler combination, obtained from measured thresholds in otologically normal young adults. The earphone is *in* the measurement — that's why the values are tabulated per transducer, and why ISO 389 has a separate part for supra-aural (Part 1, TDH-39), circumaural (Part 8, HDA200), insert (Part 2), etc. University of Manchester

Here's the reframe that dissolves the confusion: **if the coupler were a perfect ear-analog, RETSPL would be one universal curve** and there'd be no need for separate tables per transducer. The fact that the standard publishes *different* RETSPLs for TDH-39 vs HDA200 vs ER-3A is the empirical admission that the coupler is only *approximately* an ear — and that the residual approximation error is transducer-specific. So "the coupler is good enough" and "RETSPL is transducer-specific" aren't in tension. The second is the *price* of the first being only approximate. The leftover is exactly the ΔRECD term from before.

The matched-pair rule

`HL = SPL − RETSPL` is a subtraction of two coupler SPLs. It's only valid if **the same earphone produced both**. Look at what you actually have on each side:

- SPL term: produced by your **DT250** (your dB FS→SPL cal, in the 4153). ✓ DT250 coupling.
- RETSPL term (the graph): produced by the **TDH-39**. ✗ TDH-39 coupling.

You're not "continuing the same assumption." You're trusting the coupler *the same amount on both sides* — but putting a different headphone on each side and subtracting. That's the error. Not too much faith in the coupler; a transducer mismatch inside the difference.

Where your intuition is exactly right

If you measured the 0-dB-HL reference **through your own DT250** — i.e. median normal threshold on the DT250 in your 4153, which is biological calibration — then both sides carry DT250 coupling, the subtraction is clean, and you'd never need anyone's published table. Your instinct that "the coupler already bakes in the real coupling" is correct *for a curve you measure yourself*. The graph just bakes in someone else's headphone's coupling.

How wrong is the current table? Concretely.

HDA200 (circumaural, IEC 60318-1 — your coupler family) vs your current TDH-39 (supra-aural, 6 cc):

Freq	TDH-39 (current)	HDA200	Δ baked into your HL
125	45.0	30.5	**+14.5**
250	25.5	18.0	+7.5
500	11.5	11.0	+0.5
1000	7.0	5.5	+1.5
1500	6.5	5.5	+1.0
2000	9.0	4.5	+4.5
3000	10.0	2.5	**+7.5**
4000	9.5	9.5	0.0
6000	15.5	17.0	−1.5
8000	13.0	17.5	−4.5

HDA200/DD450 values from the RadioEar DD450 datasheet, IEC 60318-1 coupler with type 1 adaptor, RETSPL per ANSI S3.6-2018 and ISO 389-8:2004. Since RETSPL is the larger number subtracted, a +14.5 dB error at 125 Hz means your current setup reports low-frequency thresholds **~14.5 dB better than reality** — a normal ear would show an implausibly good (negative) 125 Hz HL. If you have any pilot audiograms, that low-frequency over-optimism is the fingerprint. This is *not* "good enough even for disclosed research" at the low end; it's a shape distortion, worst exactly where supra-aural leakage and the 6 cc coupler diverge most from a circumaural cup. So this swap isn't cosmetic — it's the difference between a believable and an unbelievable low-frequency audiogram. Interacoustics

ASAP path — and the cost point you may be missing

You do not need to buy anything to fix this today. The RETSPL table is a published reference; adopting the HDA200 values is just different numbers in `calibration_config.json`. It costs nothing and:

- fixes the *coupler* mismatch entirely — your 4153 *is* the IEC 60318-1 ear the HDA200 table assumes (the Tympan/Open Audio HDA200 RETSPL work used the same B&K 4153 + 4192 chain);
- shrinks the *earphone* mismatch from supra-aural-vs-circumaural (worst case) to circumaural-vs-circumaural (the small residual ΔRECD).

The DD450 purchase buys you something different: making your *hardware* literally the reference transducer, so ΔRECD → 0 for real. You don't need that for disclosed research. And it's moot for sourcing numbers anyway — the DD450 is designed to replicate the HDA200 and has the same RETSPL values (ANSI S3.6-2018), so the DD450 datasheet *is* your citable

HDA200 table. (One caveat: a recent re-derivation suggests a small DD450-specific correction, mostly ~5 dB at 8 kHz — but that's well inside your DT250-vs-HDA200 residual, so ignore it for now.) ResearchGate

Concretely: replace the `retspl` block's `source` and `table` in each profile with the HDA200 values above, and write the assumption into your CSV header / methods, something like: *"dB HL via ISO 389-8 / ANSI S3.6-2018 HDA200 RETSPL (IEC 60318-1). DT250 is not the reference transducer; residual transducer error (ΔRECD, DT250 vs HDA200) is unquantified. Research display only; not for clinical diagnosis."*

Two decisions before I'd touch the config:

1. Store the HDA200 table per-profile (current structure) or as one shared `retspl_hda200.csv` referenced by all profiles? Shared = one-file swap when you later do biological cal or buy the DD450.
2. Do you want me to also add the intermediate frequencies the HDA200 table defines (750, 1500 already there; also 1250, 1600, 3150 if you ever test them), or keep it to the 10 audiometric points you currently use?

> one thing i want to push back on before moving forward: i don't see any mention of the TDH-39in the figure. as far as i can tell, it's not specific to headphones at all

Reconciling Manchester source data on TDH-39 specifications >

Fair catch — the figure itself doesn't say TDH-39, and that textbook presents it as "Threshold Sensitivity of the Normal Ear," full stop. Reading it as a generic normal-ear curve is the natural reading, and I asserted the transducer provenance without flagging that the label is mine, not the figure's. Let me verify the identification rather than hand-wave it.

Corroborated transducer identity through literature and internal curve evidence >

You're right that the figure doesn't label it, and I should be clear: "TDH-39" is *my* identification of those numbers, not something printed on Roeser & Clark. So let's test the identification rather than assert it — and it holds up two independent ways, one of which needs no appeal to any label at all.

The numbers identify the transducer. That 125 Hz = 45.0 dB SPL value is the published TDH-39 supra-aural RETSPL — for TDH-39 earphones, 0 dB HL at 125 Hz corresponds to 45 dB SPL (a 70 dB HL dial reading expects 115 dB SPL), and the rest of your set (250/25.5, 500/11.5, 1000/7.0, 2000/9.0, 4000/9.5…) is the same TDH-39 column. More to your actual point: the standard *defines* these as transducer-specific by construction. RETSPLs are established according to the type of earphone, coupler, and test frequency, and ISO 389-1 states its values in terms of the response of standard models of earphone measured on a coupler. The quantity *is* "this earphone on this coupler at normal threshold" — there's no transducer-neutral version of it. That's why the standard has separate tables for TDH-39, HDA200, ER-3A inserts, even EarPods. University of Manchester + 2

The curve betrays its transducer even with the label stripped off. This is the part you can check yourself, no citation required:

- *125 Hz = 45 dB SPL.* A genuinely transducer-neutral normal-ear threshold at 125 Hz is ~22 dB SPL in a sound field, ~28–30 in a circumaural or insert coupler. 45 is ~15–20 dB higher than any of those. No normal ear *needs* 45 dB at the eardrum to detect 125 Hz — that excess is supra-aural cushion leakage, which is exactly the headphone the curve was measured with. A transducer-neutral curve physically cannot show 45 there.
- *Shape inversion at 3–4 kHz.* Your curve has 3 kHz (10.0) and 4 kHz (9.5) *less* sensitive than 1 kHz (7.0). A real ear in a sound field is *most* sensitive at 3–4 kHz — ear-canal resonance pulls the threshold below 0 dB SPL there. The fact that your curve rises instead of dipping is the fingerprint of headphone presentation: the ear-canal/concha resonance gain is bypassed, and the coupler's own response takes over. A transducer-neutral "normal ear" curve would dip at 3–4 kHz; this one bumps up.

So the curve is not "the normal ear, full stop." It's normal-ear-threshold *fused with* TDH-39-on-coupler, and the two factors can't be separated by inspection — but the supra-aural one is clearly present (the 45, the 3–4 kHz inversion).

Why this is unavoidable, not a quirk of this figure. 0 dB HL is a *perceptual* anchor — the median normal threshold. To express that anchor in dB SPL you have to specify where you measure the pressure and how the sound got to the ear, and "how it got there" is the transducer. Same perceived threshold, different delivery → different SPL at the measurement point. So *any* table mapping HL to SPL is implicitly an earphone+coupler table. The textbook's "Threshold Sensitivity of the Normal Ear (0 dB HL)" caption is pedagogical shorthand that hides the second factor — which is precisely what made the curve look transducer-neutral to you. Reasonable reading; the label just isn't telling you the whole story.

Net effect on the decision: unchanged. Subtracting this curve from your DT250 SPL still puts a supra-aural earphone on the reference side and your circumaural one on the measurement side. The HDA200 table (circumaural, matched to your 60318-1 coupler) remains the right swap — and it's the same physics that explains *why* the HDA200 125 Hz value (30.5) is ~14 dB below this one: less low-frequency leakage.

ok, that helps to clarify things... but you said "No normal ear needs 45 dB at the eardrum to detect 125 Hz." can you provide a table of what a "normal ear" needs to detect sound at specific frequencies? i think that could be the missing link to solve this, if we stick with the assumption that the artificial ear = reality

Unearthing ISO standards clarifying threshold discrepancies >

Good question — and getting the right table is genuinely illuminating, but I want to be upfront: I think this curve will *clarify* the problem rather than dissolve it, for a specific reason. Let me pull accurate numbers first.

Synthesized threshold data across transducers and reconciled ISO standards >

This is the right question to chase down, and the table is genuinely clarifying — but the punchline is that the curve you're hoping for *does* exist, and the standards body explicitly tells you not to use it the way you want to. Here's the three-way comparison at your audiometric frequencies:

Freq (Hz)	Free-field "normal ear" (ISO 389-7)	Circumaural HDA200 (ISO 389-8)	Supra-aural TDH-39 (your current)
125	22.1	30.5	45.0
250	11.4	18.0	25.5
500	4.4	11.0	11.5
1000	2.4	5.5	7.0
1500	2.4	5.5	6.5
2000	−1.3	4.5	9.0
3000	−5.8	2.5	10.0
4000	−5.4	9.5	9.5
6000	4.3	17.0	15.5
8000	12.6	17.5	13.0

Free-field values: ISO 389-7:2019, frontal-incidence free field — 125 Hz = 22.1, 1 kHz = 2.4, 3 kHz = −5.8, 4 kHz = −5.4, 8 kHz = 12.6 dB SPL. iTech

The first column is the literal answer to "what does a normal ear need": the SPL of a frontally-incident free-field tone at threshold, measured where the centre of the listener's head would be, with the listener absent, listening binaurally. That's as transducer-neutral as it gets. ISO

And here's the catch, straight from the standard. ISO 389-7 carries an explicit warning: these threshold data differ from the audiometric zero in ISO 389-1/-2/-5/-8 because those refer to monaural listening through earphones with SPL referred to specified couplers and ear simulators, so direct comparison is not appropriate. In other words: you cannot subtract the free-field column from your coupler-measured SPL. ISO says so by name. ISO Iteh

Why not — look at the 3–4 kHz rows. Free-field threshold goes *negative* there (−5.8, −5.4 dB SPL). That's not the cochlea being magically sensitive; it's your outer ear. Head diffraction + concha + ear-canal resonance add ~10–15 dB of acoustic gain around 3 kHz, so the *field* only needs to be below 0 dB SPL because your ear amplifies it before it reaches the drum. That gain is the HRTF. A headphone sits *past* the outer-ear amplifier — it bypasses the concha/canal resonance — so it gets no free gift, and its RETSPL has no dip (HDA200 = 2.5, TDH-39 = 10.0 at 3 kHz, both well above the free-field −5.8). Subtracting the free-field curve from a headphone measurement would falsely credit your DT250 with ~8–15 dB of outer-ear gain it never delivered, and reintroduce a 3–4 kHz notch that isn't there. (Plus free-field is binaural and earphone refs are monaural — a smaller mismatch on top.)

Now your actual lever: "if we stick with artificial ear = reality." Follow that assumption all the way through. "Artificial ear = reality" means coupler SPL = real-eardrum SPL. Granting it, the curve you'd recover at threshold is the *eardrum* threshold — and that is still **not** the free-field "what the ear needs" curve (they differ by exactly the HRTF above), and it is still transducer-dependent, because clamping a headphone onto the ear changes the threshold itself: the seal traps low-frequency physiological noise and alters the ear-canal load, by different amounts for a leaky supra-aural cushion vs. a sealed circumaural cup. That's most of why the 125 Hz column runs 22.1 → 30.5 → 45.0 as you go field → circumaural → supra-aural. So even with your assumption granted in full, there is no single transducer-neutral curve to subtract. The "missing link" is missing because the thing it would have to be — a threshold that's both eardrum-referenced *and* independent of what's on the ear — isn't a real quantity.

The satisfying part: your assumption, applied correctly, lands exactly on the HDA200 column. "What a normal ear needs, measured in this artificial ear, when sound is delivered by a circumaural earphone" *is* the ISO 389-8 RETSPL. It's the eardrum-referenced, coupler-matched, circumaural version of the free-field curve — the legitimate embodiment of "artificial ear = reality" for your hardware class. Your DT250 is circumaural in the same IEC 60318-1 ear the HDA200 table assumes, so the residual is just DT250-vs-HDA200. So your instinct ("trust the coupler, subtract a normal-ear curve") is right — it just resolves to the middle column, not a universal one, and definitely not the free-field one or your current supra-aural one.

So the table doesn't unlock a transducer-free shortcut, but it does settle the choice: of the three columns, the middle one is the one that matches both your coupler and your earphone type. Same swap as before, now with the full picture behind it.

ok, interesting. and how about the fletcher-munson equal loudness curves, could those help?

Examined equal-loudness curves, identified frame mismatch problem >

Same answer as the free-field curve — because the equal-loudness contours *are* that curve, plus a lot of information that's irrelevant to HL. It's the natural next thing to reach for, so let me dismantle it properly rather than just say no.

The lineage matters here. Fletcher-Munson (1933) → Robinson-Dadson (1956) → ISO 226 (2003, revised since) are all the same construct: equal-loudness contours measured in a free field, frontal incidence, binaural. The bottom contour — the limiting low-level one, which ISO 226 also tabulates explicitly as the threshold-of-hearing function — is essentially the same curve as the free-field column I put in that table. ISO 389-7 and ISO 226 are harmonized and draw on overlapping data. So the threshold contour inherits the identical frame mismatch: the HRTF is baked in (that's the 3–4 kHz dip going negative), it's binaural, and it's referenced to free-field SPL where your head would be, not coupler SPL at the eardrum. You can't subtract it from a headphone measurement for exactly the reasons you couldn't subtract the free-field column.

Fletcher-Munson *specifically* picks up two extra strikes. First, it's the 1933 original, measured over the earphones of that era and known to be inaccurate, especially at low frequencies — which is why it was superseded twice. Nobody uses the literal F-M curves metrologically anymore. Second, because it was earphone-based, it isn't even transducer-neutral: it's tied to 1933 earphones with their own coupling. So it's the worst of both worlds — transducer-specific to a transducer you can't obtain, *and* obsolete.

But the deeper issue is conceptual, not provenance. HL is a threshold-referenced scale: 0 dB HL is the median normal threshold, and `dB HL = dB SPL − RETSPL` is a pure per-frequency offset. The defining content of equal-loudness contours — that the ear's frequency response *changes with level*, so the contours aren't parallel; they flatten as SPL rises (this is precisely why A-weighting, built on the ~40-phon contour, rolls off the lows hard while C-weighting, near 100 phon, is nearly flat) — is suprathreshold *loudness* information. HL doesn't use any of it. To anchor 0 dB HL you only ever need the single bottom contour, which is the threshold, which is the free-field curve you already rejected. The rich part of the equal-loudness data is irrelevant to your problem, and the part that's relevant is the frame-mismatched part.

Equal-loudness contours do earn their keep elsewhere — loudness modeling, deriving frequency weightings, suprathreshold loudness balancing, possibly your LDL work. Just not the HL threshold anchor.

So they route back to the same destination. Free-field threshold, equal-loudness contours, MAP — they all fail to substitute for one root reason: none is referenced to your coupler-and-circumaural-earphone frame. The only curve that is, is a coupler-referenced circumaural threshold reference, i.e. the HDA200 RETSPL. That remains the answer; every neutral-curve detour you try will circle back to it because the neutrality you want is the specific thing that disqualifies them.

> ok, so given everything we've discussed, can you clearly map out the best strategy to produce an accurate dB HL conversion (and how the current implementation needs to be updated to achieve it), given the constraints/goals that :
> a) we want to use what we already have (headphones, artificial ear), rather than buy new headphones
> b) our situation is not for clinical use, so we can target "good enough", particularly if we can clearly describe/justify assumptions we've made and their impact

Reconciling PRD framework with strategic planning request >

This is a planning request, so before writing the spec let me check the project's house format for feature PRDs.

Cataloging paradigms and reconciling dB HL conversion strategies >

Read it. This is now a plan-and-approve step (no code edits yet). Before I write the PRD, one scope check the skill flags — which paradigms currently emit an HL column, so the hardening change doesn't silently drop one.

Crystallizing strategy framework, awaiting user approval before proceeding >

Useful — that catches a real one. Three paradigms currently emit an HL column (`level_units = "dB SPL"`): PTA, LDL, **and SRT**. PTA and LDL are legitimate (single-ear dB SPL level at a pure-tone fc). SRT is not: it's a *speech* threshold, which in HL terms needs a speech RETSPL (per IEC 60645), not the pure-tone RETSPL(fc) the current code applies. So SRT is quietly producing a wrong HL value. The hardening change fixes that too. Writing the PRD now.

Orchestrated strategic reframe with tiered plan and implementation checkpoint >

PRD's in the file. The headline, so you don't have to dig for it:

The strategic reframe: you're not stuck approximating a transducer you don't own. You *have* the HDA200 (uncalibrated) — that's the actual reference transducer. So the plan is two-tiered:

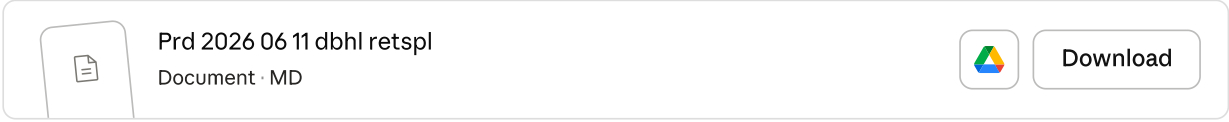
- **Tier 0 (ship today):** swap the TDH-39 table → ISO 389-8 HDA200 values. Pure config change; the loader already builds the converter from the table. Disclosed assumption: ΔRECD(DT250,HDA200) ≈ 0. Unblocks you immediately and kills the ~14.5 dB low-frequency error you have now.
- **Tier 1 (the accuracy target, existing hardware):** calibrate the HDA200, then run a small normal-hearing panel on DT250 *and* HDA200 in the same session. The within-subject difference gives you DT250-specific RETSPL directly — this is the only way to actually recover ΔRECD (it's a real-ear quantity, invisible to the coupler), and because it's a *difference* across the same ears, ~5–10 ears is enough rather than the large panel an absolute biological cal would need.

One bug fell out of the scope check: SRT currently emits a `threshold_dB_HL` computed with the pure-tone RETSPL(fc) — wrong for a broadband speech threshold (speech needs its own RETSPL). The plan replaces the fragile implicit gate with an explicit `hl_reportable` flag: PTA + LDL on, SRT off, everything else unchanged.

The implementation itself is small and architecture-preserving — your SPL-canonical/HL-derived design doesn't change. Code touched: `calibration_config.json` (data), `adaptive_paradigm.py` (one gate), `paradigm_pta/ldl/srt.py` (one flag each), optional GUI label.

Six open questions at the end of the PRD — the load-bearing ones are #1 (commit to Tier 1 now, or Tier 0 only?) and #2 (OK to suppress SRT's invalid HL now?).

Not writing any code until you've approved this. And when we do: upload fresh local copies of those five files first — the `/mnt/project/` snapshot is already behind your local SPL numbers, so I'd be editing stale versions otherwise.



on additional detail to consider: the team has confirmed that PTA is the only threshold that needs to implement dB HL

please adjust the PRD accordingly

Recalibrated PRD scope to restrict dB HL reporting exclusively to PTA >

Got it — that simplifies things and makes the explicit-flag approach cleaner (only PTA opts in; LDL and SRT both lose their currently-emitted HL columns). Let me update the PRD. Viewing it first to edit precisely.

Finalizing edits and presenting updated PRD >

Updated. The PTA-only decision actually tightens the design — the explicit flag now encodes team policy directly, with PTA the sole opt-in.

What changed:

- **PTA** → `hl_reportable: True` (sole HL paradigm).
- **LDL** → `hl_reportable: False`. This *removes* a column LDL currently emits. LDL HL was technically valid (level at fc) but is now out of scope — flagged explicitly so it's a documented decision, not a silent drop.
- **SRT** → `hl_reportable: False`. Drops it for two reasons now: out of scope *and* it was invalid anyway (pure-tone RETSPL applied to a speech threshold).
- Everything else unchanged.

The RETSPL table swap and the gate change are untouched — narrowing to PTA doesn't affect them, since the RETSPL table is profile-level calibration data consumed only when HL is computed (now only by PTA).

One thing worth your eye in the open questions: dropping LDL's HL column is the only place this touches *current* output behavior, so #2 is really "confirm we're OK losing LDL's HL column." Everything else is additive or a fix.

Still holding for approval before any code — and a fresh upload of the five files when you're ready to implement, since the snapshot here lags your local SPL numbers.

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ok, one more question: it seems that the ISO 389-8 only goes from 125 Hz to 8 kHz, does that mean we can't generate dB HL values for frequencies above 8k?

Investigated ISO 389-5 standards and EHF reliability limitations >

Good question, and the answer has a practical fork — let me check whether the battery actually reaches up there, then sort out the standards picture.

Clarifying ISO 389-8's frequency range and coverage limitations >

The battery itself answers most of this: every defined frequency set tops out at 8 kHz (`AUDIOMETRIC_FREQS` and `EXTENDED_FREQS` both end at 8000; PTA's default `fc` is 500–4000). So for everything you actually test today, ISO 389-8 covers you completely — the 8 kHz ceiling isn't a live constraint. Let me confirm the extended-high-frequency standards picture for the forward-looking case.

Reconciled ISO standards coverage and identified EHF measurement limitations >

Confirmed the picture, and it splits cleanly into "not a problem now" and "here's the story if you extend."

For what you test today: no issue. Your frequency sets stop at 8 kHz, and ISO 389-8 covers 125 Hz–8 kHz, so every PTA frequency you can currently select has a valid RETSPL. The 8 kHz ceiling of 389-8 isn't constraining you.

If you extend into extended-high-frequency (EHF) territory: the gap is filled by a *different* standard, not left open. ISO 389-5:2006 specifies RETSPLs for pure tones from 8 kHz to 16 kHz for specific circumaural and insert earphones — the HDA200 is one of them, and the standard was reconfirmed as current in 2025. So you'd just extend the table and cite ISO 389-5 for the >8 kHz rows. The HDA200/DD450 EHF values (from that lineage) run roughly: 9k → 19, 10k → 22, 11.2k → 23, 12.5k → 27.5, 14k → 35, 16k → 56 dB SPL. Above 16 kHz there's no ISO audiometric zero at all — research studies go to 18–20 kHz, but you'd be defining/borrowing a reference and disclosing it. Iteh iso

But EHF HL is much softer than conventional, for stacked reasons — worth knowing before anyone reads those numbers as comparable to your ⩽8 kHz audiogram:

1. The EHF metrology is explicitly provisional. The EHF references cover only a small number of specific earphones, aren't age/gender specific, and rest on 18–25-year-old subjects — there's active research precisely because EHF norms are unsettled. ResearchGate
2. The coupler is itself interim. ISO 389-5 uses the IEC 60318-1 ear simulator with an IEC 60318-2 adapter as an interim acoustic coupler for EHF circumaural calibration. Your B&K 4153 is the 60318-1 ear; unless you also have the 60318-2 EHF adapter, your *own SPL calibration* above 8 kHz wouldn't even be in the matching coupler condition — so both sides of `SPL − RETSPL` degrade. Iteh
3. The DT250-vs-HDA200 mismatch you're already accepting gets *worse* at EHF, not better. Up there the wavelength approaches coupler/ear-canal dimensions, so standing waves, placement, and individual ear-canal length dominate — ΔRECD balloons and varies a lot ear-to-ear.
4. Plus the obvious: DT250 reproduction and your inverse-filter accuracy at 10–16 kHz are their own question.

One implementation gotcha regardless: don't lean on the `HLConverter`'s extrapolation above 8 kHz. It extrapolates log-linearly from the top two points (your 6k→8k slope), but real EHF RETSPL climbs steeply and nonlinearly (8k=17.5 → 16k=56). Log-linear would badly under-predict. The "extrapolating; HL values here are unverified" warning it emits is correct and should stay — and if you ever go EHF, add explicit ISO 389-5 rows rather than trust the extrapolation.

No PRD change needed for current scope (your table = the battery's range). If you want, I can add one line to the PRD's Out-of-scope — "EHF (>8 kHz) HL: ISO 389-5 table + 60318-2 adapter required; extrapolation not trusted" — so the boundary is on record. Say the word and I'll drop it in.

